

Advanced (Habanero)

Q1. Simplify $\sqrt{25}$

- (A) 5
- (B) -5
- (C) ± 5
- (D) 25
- (E) $-\sqrt{25}$

Q2. Simplify $\sqrt{64}$

- (A) 8
- (B) -8
- (C) ± 8
- (D) 64
- (E) $-\sqrt{64}$

Q3. Simplify $\sqrt[3]{27}$

- (A) 3
- (B) -3
- (C) ± 3
- (D) 27
- (E) $-\sqrt[3]{27}$

Q4. Convert to radical form: $x^{1/2}$

- (A) $\sqrt{x}$
- (B) $\sqrt[3]{x}$
- (C) x^2
- (D) $x^{1/3}$
- (E) $-\sqrt{x}$

Q5. Convert to rational exponent form: $\sqrt[3]{x}$

- (A) $x^{1/3}$
- (B) $x^{1/2}$
- (C) x^2
- (D) $x^{2/3}$
- (E) $x^{3/2}$

Q6. Simplify $\sqrt{16x^2}$

- (A) $4x$
- (B) $4|x|$
- (C) $-4x$
- (D) $-4|x|$
- (E) $4|x|$

Q7. Simplify $\sqrt[3]{-8x^3}$

- (A) $-2x$
- (B) $2x$
- (C) $-2|x|$
- (D) $2|x|$
- (E) $-2x^2$

Q8. Simplify $\sqrt{x^2 + y^2}$

- (A) $x + y$
- (B) $x - y$
- (C) $|x + y|$
- (D) $\sqrt{x} + \sqrt{y}$
- (E) $|x| + |y|$

Open Ended Questions (FRQ)

Q9. Simplify the expression $\sqrt{72}$ and express the answer as a product of a whole number and a simplified radical.

Q10. Simplify $\sqrt[3]{-64x^3}$ and express the answer as a product of a whole number and a simplified radical.

Chef's Tips

- Read each question carefully
- Show all work and simplify answers
- Check units and sign of the answer